

Operational Efficiency: Automated data extraction, transformation, and loading (ETL) processes reduce manual data entry and manipulation efforts, saving time and minimizing errors.

Strategic Planning and Budgeting: Accurate and comprehensive data supports effective strategic planning, budgeting, monitoring and allocation of resources for future growth and initiatives.

Effective Patient Care: Hospital staff can access and utilize a unified dashboard related to patient care, cost of treatment, patient experience and staffing utilization which aiding in care optimization and efficiency.

Unified Reporting and Analysis: Sidra executives, clinical and research staff can access integrated reports and dashboards that provide a holistic view of operations, finances, patient care, and other critical aspects.

Enhanced Performance Measurement: Key performance indicators (KPIs) can be monitored consistently, allowing the hospital to assess its performance and align activities with strategic objectives.

Reduced Data Latency: Real-time or near-real-time data updates allow for faster reporting, making critical information available for time-sensitive decisions.

Self-Service Analytics: Hospital staff can use user-friendly tools to perform ad hoc queries and analyses without needing extensive technical skills, promoting self-service data exploration and analysis.

Data Visualization: Interactive visualizations and dashboards provide intuitive ways to communicate complex data, enabling quicker understanding of trends and insights.

Data Governance and Security: The Data-Lake allows for centralized data governance, ensuring data security, privacy, access control, and compliance with regulatory standards and best practices (i.e. Role Based access control and data encryption).

Data Catalog and Metadata Management: The data-lake can provide a centralized repository to catalog and manage the metadata associated with the stored data.

Long-Term Data Retention: The data-lake provides a secure and accessible repository for historical data, ensuring valuable insights are retained even as ERP systems evolve.

Predictive Analytics, AI, and Machine Learning: The data-lake solution can provide a rich dataset and platforms for training predictive models, AI, and machine learning algorithms, and support data science workloads. It includes tools and frameworks for statistical analysis, machine learning, and AI.

- Address any issues or concerns that arise during deployment.
- Deliverable: Production Data Warehouse and Analytics Environment

3.12 Post-Implementation Support

- Provide ongoing technical support to address any system issues, bugs, or enhancements.
- Monitor system performance, data quality, and user interactions.
- Implement updates and modifications based on evolving business requirements.
- Deliverable: Support and Maintenance Documentation

4. Software Licenses

The vendor should be able to develop and implement the solution using a software stack currently available at Sidra Medicine without additional licensing cost. Please refer to the below technology stack available:

- Azure Databricks
- Azure Data Factory
- Azure Synapse Analytics
- Microsoft PowerBI
- Azure Machine Learning or Databricks Mosaic ML
- Other Azure platforms available within Azure Qatar data center.

If additional software licenses are required, it needs to be mentioned clearly with the projected licensing cost.

5. Timeline and Milestones:

The project is estimated to be completed within 10 months following the major activities and deliverables listed in section 3 above.

The vendor should provide a detailed project plan indicating the project management methodology they will follow.

Activity/Deliverable	Delivery Schedule (from start of engagement)
Requirements gathering and analysis	Month 1
Data-Lake / Data-Lakehouse architecture design	Month 1
Infrastructure setup	Month 1 – Month 2
Data ingestion and integration (both batch and real-time).	Month 1 – Month 5
Data processing and transformation	Month 2 – Month 7

Technical Expertise:

- Deep knowledge of Data-Lake technologies in general and specifically Azure Databricks and Azure Data Factory stack.
 - Proven expertise in developing and deploying large scale Analytics projects.
 - Deep knowledge and proven expertise in Data Architecture and ETL/ELT best practices, implementation, optimization and support.
 - Deep knowledge proven expertise in Data Governance standards and best practices.
 - Experience in visual analytics and dashboards development using PowerBI platform.
 - Vendor should be listed as Microsoft and Databricks official partner (Gold Partner preferred).
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- Proposal should be very clear and self-explanatory with a straightforward, concise description for each component/deliverable as listed in this RFP request.
 - Delivery mode for the project shall be hybrid with on-site technical engineers and resources throughout the project implementation and post go-live support.
 - Proposal should provide all the technical details related to Data Architecture, ETL/ELT technologies and ML platforms.
 - The proposal should clearly define how the platform will be scalable to handle increased data volume and user loads in the future as well as accommodate new data sources and systems.
 - The proposal should define clear metrics that will be used for data processing, storage and retrieval times.
 - The vendor must have done similar implementations in other organizations
 - The vendor must provide proper references and sample reports for such implementations.
 - The vendor must provide all the documentations required as stated in the deliverable section.
 - The vendor must comply with all technical requirements and information security guidelines.